

Magnitude, Not Shape: Volatility Beats Downside- and Tail-Risk Metrics at Forecasting Drawdown

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ABSTRACT

Every few years a new downside- or tail-risk metric promises to improve on plain return volatility. The premise is always the same. The *shape* of the loss distribution, whether its asymmetry, tail heaviness, or co-crash structure, is supposed to carry information that symmetric volatility misses. We test that premise as adversarially as the data allow. On an evaluation that is survivorship-free where survivorship can be fixed (crypto), multi-regime, and multiple-testing-corrected, with survivorship-biased equity beds as a cross-asset check, no metric reliably beats volatility at forecasting an asset’s forward maximum drawdown across ten metrics and five test beds, whether the metric is published or machine-discovered. The failure has a clean anatomy, which we call *magnitude, not shape*. The only increments that survive Benjamini–Hochberg correction come from metrics that embed the *size* of downside losses (Value-at-Risk on all five beds, reaching 0.19 at its largest; downside deviation; and, weakly, negative semibeta). Every magnitude-free shape metric adds nothing or subtracts. Yet the shape metrics are not broken. On genuinely shape-sensitive targets they beat volatility. Drawdown is simply a magnitude phenomenon that shape does not drive. Volatility is hard to beat because it is persistent, though a predictive mediation test indicates it is not a sufficient statistic. The tiny magnitude edge does not pay its way either. A cost-aware backtest and a pre-registered composite add nothing over volatility outside fat-tailed crypto. Finally, an autonomous large-language-model discovery loop only inflates validation while its locked-test score plateaus at volatility, a data-snooping caution now that agentic search makes signal-hunting nearly free. On a level field, plain volatility is remarkably hard to displace.

CCS CONCEPTS

• **Computing methodologies** → **Machine learning**; • **Applied computing** → **Economics**.

KEYWORDS

downside risk, tail risk, drawdown, volatility, survivorship bias, multiple testing, data snooping, LLM agents

1 INTRODUCTION

Prices fall faster than they rise, and it is the fall that ruins investors. From that intuition a large literature has grown. It proposes metrics that look past symmetric volatility to the shape of the loss distribution. These include lower partial moments and the Sortino ratio [6, 14, 23], the partial-moment “predictive” ratio (UPM/LPM) [25], left-tail Value-at-Risk (VaR) and expected shortfall (ES) [3, 22], realized semibetas [8], downside beta [2, 18], lower-tail dependence [9], disappointment-aversion risk (GDA) [13], and tail-exponent factors [17]. Each rests on the

same premise, namely that asymmetry or tail structure carries information volatility misses.

We test that premise on the fairest field we can build. Three disciplines the source literature usually omits turn out to matter enormously. A *survivorship-free* universe keeps the assets that actually died. A *locked* out-of-sample holdout stops search from leaking into the verdict. Finally, *multiple-testing correction* keeps an evaluation of dozens of comparisons from manufacturing winners. We evaluate ten metrics, volatility and nine downside- and tail-risk challengers, as forecasters of 90-day forward maximum drawdown across five cross-sectional test beds spanning calm, bull, and crash regimes in crypto and equity, with per-date rank correlations, paired block-bootstrap confidence intervals, and Benjamini–Hochberg (BH) correction.

The headline is negative and, in hindsight, unsurprising. No metric reliably beats plain volatility. Our contribution is the anatomy of that result and the integrity of the evaluation behind it. First, a clean empirical law we call *magnitude, not shape*. The only signal surviving correction beyond volatility is the *size* of tail losses (VaR/ES/drawdown deviation), not their asymmetry (§4). Second, volatility is hard to beat because it is persistent, though a predictive test indicates forward volatility is not a sufficient statistic (§5). Third, the residual edge does not pay its way, except in fat-tailed crypto (§5). Fourth, an autonomous large-language-model (LLM) discovery loop does not beat volatility out-of-sample and only overfits validation, a data-snooping caution as agentic search makes signal-hunting nearly free (§5).

Related work. Lower partial moments generalize semivariance [6, 14] and underpin the Sortino ratio [23] and the partial-moment ratio [25]; tail measures include VaR/ES [3, 22], semibetas [8], downside beta [2, 18], lower-tail dependence [9], disappointment aversion [13], and tail exponents [17]. Most were built to price the cross-section, not forecast one asset’s drawdown; a separate literature studies the drawdown statistic directly [10, 20]. Volatility itself is a persistent, tradable risk signal [15, 21]. That volatility is persistent is the founding fact of ARCH/GARCH [1, 7, 11, 12]; that searching many signals inflates in-sample rank correlations is classical data snooping [4, 5, 16, 24, 28]. A recent wave of LLM-driven and agentic systems automates signal discovery in finance [19, 26, 27], and typically reports validation-selected wins. Our contribution is to follow such wins onto a locked test. An *autonomous* agent reaches the snooping frontier at near-zero cost, and its edge evaporates once a survivorship-free holdout is scored exactly once.

2 EXPERIMENTAL SETUP

Five test beds. *Crypto (survivorship-free)*: a CoinMarketCap daily panel (2013–2018) that retains crashed and delisted coins, so a point-in-time top-100-by-market-cap universe includes assets that later crater; we use three regimes: 2016 (calm), 2017 (bull),

2018 (crash). *Equity*: S&P 500 constituents, 1994–1999 (calm) and 2005–2009 (the Global Financial Crisis). The equity beds are survivorship-biased by construction (freely available equity data retains almost no true deaths, itself a telling finding), so the survivorship claim rests on the crypto beds, while equity tests cross-asset generality. We checked this directly. A public price dump reputed to include delisted stocks (the Kaggle Huge Stock Market dataset) contains none of our windows’ famous S&P 500 casualties, with Lehman, Bear Stearns, Enron, and WorldCom all absent, and yields zero mid-window delistings, so we could not source a genuinely survivorship-free equity bed from free data. The bias runs in our favor. On crypto-2018, dropping the coins that died flattens shape and inflates the magnitude edge (§6), so the small equity increments are plausibly upper bounds and shape’s equity penalty a lower bound.¹

What we forecast. From a 180-day trailing window we forecast each asset’s 90-day forward maximum drawdown, coding a delisting as a -100% loss. We score a metric by its per-date cross-sectional rank correlation with that drawdown, averaged over the bed, cross-sectional rather than time-series because the operational question is which assets to de-risk on a given date. The beds span 8, 9, and 7 formation dates on crypto-16/17/18 and 66 and 54 on equity-94/05, each ranking a point-in-time cross-section of up to 100 crypto assets or the ~ 500 S&P 500 constituents.

Metrics. All ten are computed from the same 180-day trailing daily returns. Volatility is the return standard deviation. Three are tail-magnitude measures: historical 5% Value-at-Risk and expected shortfall [22], and downside deviation, the lower partial moment of order two about a zero target [6, 23]. The rest are tail-shape measures: the Viole–Nawrocki upper/lower partial-moment ratio [25]; a per-asset Hill loss-tail exponent, our per-date simplification of the Kelly–Jiang common tail factor [17]; and four systematic co-movement measures, namely negative realized semibeta [8], downside beta [2], lower-tail dependence [9], and a disappointment-aversion proxy [13], each estimated against the equal-weight market. The partial rank correlation $\rho(\cdot, \text{drawdown} \mid \text{volatility})$ is the Spearman correlation of the two residual ranks after regressing the metric and the drawdown on volatility, per date.

Significance and power. We attach a paired block-bootstrap 95% confidence interval (CI) to every comparison, using length-3 blocks over the per-date series (near the $n^{1/3}$ rule of thumb for these lengths); the VaR increment’s sign and significance are unchanged under block lengths 1–4 and a stationary bootstrap. We apply BH correction at $q = 0.05$ within each test battery (the head-to-head and the partial family, ~ 45 cells each), running BH within rather than across batteries because the two answer different questions. The crypto beds are thin (7–9 formation dates), so we read crypto-only significance as suggestive and back each crypto claim with its minimum detectable effect (MDE). VaR’s increment exceeds the 80%-power MDE on crypto-16 and crypto-17 but falls

Table 1: Standalone out-of-sample drawdown-forecast Spearman correlation. * = significantly beats volatility, † = significantly worse (both survive BH, $q = 0.05$); unmarked = indistinguishable. The Hill tail exponent [17] is omitted here (not estimable per date on equity); its increment appears in Table 2. Bold marks the highest value in each column.

Metric	cr-16	cr-17	cr-18	eq 94	eq 05
volatility (base)	0.62	0.36	0.38	0.61	0.56
downside deviation	0.62	0.38*	0.41	0.58†	0.54†
VaR	0.61	0.40*	0.36	0.58†	0.54†
ES [22]	0.62	0.35	0.33†	0.58†	0.52†
neg. semibeta [8]	0.40†	0.26†	0.31†	0.49†	0.46†
downside beta [2]	0.01†	0.09†	0.15†	0.24†	0.34†
lower-tail dep. [9]	−0.38†	−0.15†	−0.29†	−0.07†	−0.05†
GDA [13]	−0.06†	−0.11†	−0.05†	−0.24†	−0.32†
UPM/LPM [25]	−0.55†	−0.28†	−0.35†	−0.43†	−0.42†

below it on crypto-18. The equity beds are well powered. There the 95% CI on every shape increment excludes effects larger than ≈ 0.04 (the 80%-power MDE is 0.034–0.042), so “shape adds nothing” is an equivalence result, not a failure to detect. A placebo that permutes forward drawdown within each date collapses the volatility correlation to ≈ 0 on four beds (≤ 0.012) and to $+0.028$ on the thinnest, crypto-2018 (7 dates), whose bootstrap CI [0.003, 0.059] barely clears zero. Against the true $+0.38$ that is a 93% collapse, the residual a seven-date within-date permutation leaves rather than pipeline leakage.

3 THE VERDICT: NOTHING BEATS VOLATILITY

The verdict is blunt. No metric beats volatility in a way that survives a change of asset class. Table 1 ranks each metric against volatility standalone. The only metrics that significantly beat volatility on any bed are two volatility variants, downside deviation and VaR, on the single crypto-bull bed, and both reverse on equity, where volatility significantly beats them. Every genuinely different, shape-based metric is significantly worse than volatility on every bed.

We fix the bar in advance to avoid a moving goalpost. An edge counts as *generalizable* only if a metric BH-significantly beats volatility on at least one crypto and one equity bed, i.e., survives a change of asset class. No metric clears it. Small, regime-specific edges exist for volatility’s own variants, but none generalizes. We pre-registered the main tests, fixing them before we saw any results so we cannot cherry-pick them, namely the verdict, the magnitude-versus-shape increments, and the composite. The finer slices by market stress and across beds were explored afterward and count only as suggestive.

4 THE ANATOMY: MAGNITUDE, NOT SHAPE

If nothing beats volatility, what is left to forecast beyond it? The *partial* rank correlation (Spearman $\rho(\text{metric}, \text{drawdown} \mid \text{volatility})$, which isolates what a metric adds beyond volatility) answers this. Figure 1 plots it for every metric and bed, and

¹All code, data pointers, a pre-registration record (the frozen tests, the generalizability bar, and the composite), and result logs to reproduce every table and figure are available in an anonymized repository for review and will be released publicly upon acceptance. Data sources are a CoinMarketCap daily crypto dump (retaining delisted coins), yfinance S&P 500 constituents and the market index, and point-in-time index membership.

the picture splits cleanly by whether a metric embeds the *size* of downside moves. Every increment that survives correction comes from a magnitude-bearing measure. VaR is significant on **all five beds** (partial ρ up to 0.19) and downside deviation on the three crypto beds. One systematic metric joins them. Negative realized semibeta, whose down-down semicovariance construction embeds downside magnitude, carries a BH-surviving increment of +0.08, +0.13, and +0.05 on crypto-17, crypto-18, and equity-05. Every magnitude-free shape measure adds ≈ 0 or subtracts, on both sides of the own-asset/systematic divide. The two own-asset shape metrics (the Viole–Nawrocki ratio and the Hill tail exponent) are themselves ≈ 0 /negative, and so are the magnitude-free co-movement measures (downside beta, lower-tail dependence, disappointment aversion). The law is therefore sharper than *own-asset vs systematic*. An increment beyond volatility appears wherever a metric carries downside magnitude, in an asset’s own tail or in its co-crash with the market, and nowhere else. The increment is also not an artifact of the -100% delisting code. Mid-window delistings are $\leq 1\%$ of asset-dates, and valuing them at last price or dropping them moves VaR’s increment by ≤ 0.02 (Table 3). Table 2 gives the exact increments with BH significance.

VaR is worse than volatility standalone on equity (Table 1) yet has a positive increment there. As a standalone ranker it is a noisier proxy for volatility, but the partial correlation isolates its small orthogonal component. The increment is also larger where markets are fat-tailed and blow-up-prone. Crypto increments run 0.12–0.19 where excess kurtosis is 8–21, against equity increments of 0.03–0.08 where kurtosis is far lower. A pooled per-date regression of the increment on tail-fatness across all 144 dates gives a positive slope (+0.013, 95% CI [0.007, 0.021]); clustering by bed widens the CI to $[-0.004, 0.032]$ (one-sided $p = 0.10$), so we read it as suggestive. Tail magnitude carries incremental information precisely when the tail is active. When crypto dates are split at the cross-sectional median of volatility, the increment rises from +0.13 on low-stress dates to +0.18 on high-stress ones, while on equity it stays flat and small.

4.1 Is the target the reason?

Expected maximum drawdown scales with $\sigma\sqrt{T}$ [20], so drawdown is magnitude-dominated, and one might object that a magnitude metric is bound to win. The objection is correct, and it is the point. Drawdown is the loss that ruins investors, and the literature proposes shape metrics to forecast it. To confirm the shape metrics are not strawmen, we re-score every metric against genuinely shape-sensitive forward targets, namely forward return skew (sign-flipped so higher is more left-asymmetric), the downside fraction (forward downside deviation over forward volatility), and the tail-exceedance frequency (share of forward days below the trailing fifth percentile). The picture inverts (Table 4). Volatility’s rank correlation on these targets is near zero or negative, while the shape metrics beat it on nearly every bed. Shape metrics forecast shape. Shape is simply not what drives drawdown. The daily adaptations are therefore faithful rather than weakened. The Hill exponent is a per-asset simplification of the Kelly–Jiang common tail factor, and the Viole–Nawrocki ratio

Table 2: Partial Spearman $\rho(\text{metric, drawdown} \mid \text{volatility})$. Rows 1–3 are own-asset tail magnitude; row 4 (negative semibeta) is magnitude-bearing through its down-down semicovariance; rows 5–9 are magnitude-free shape. * survives BH ($q = 0.05$, positive); † survives BH negative; n/e = not estimable per date (too few loss-tail observations to fit a Hill exponent per date on equity). Bold marks the highest value in each column.

Metric	cr-16	cr-17	cr-18	eq 94	eq 05
VaR	+0.12*	+0.19*	+0.13*	+0.03*	+0.08*
downside dev.	+0.13*	+0.14*	+0.20*	+0.02	+0.01
ES [22]	+0.14*	+0.08*	+0.08	+0.03	+0.00
neg. semibeta [8]	+0.02	+0.08*	+0.13*	+0.03	+0.05*
UPM/LPM [25]	−0.00	+0.01	−0.05	−0.05†	−0.07†
downside beta [2]	−0.08†	+0.03	+0.02	+0.01	+0.04
Hill tail [17]	−0.09†	−0.13†	−0.09†	n/e	n/e
lower-tail dep. [9]	−0.07	−0.04	−0.17†	−0.01	+0.00
GDA [13]	+0.03	−0.05	+0.05	−0.01	−0.03

Table 3: VaR partial $\rho(\text{VaR, drawdown} \mid \text{vol})$ under three delisting codings. Baseline (-100%) and last-price coincide (mid-window delistings are rare); the increment is unchanged to within ± 0.02 .

Bed	baseline (-100%)	last-price	exclude
crypto-16	+0.12	+0.12	+0.14
crypto-17	+0.19	+0.19	+0.19
crypto-18	+0.14	+0.14	+0.14
equity-94	+0.035	+0.035	+0.035
equity-05	+0.076	+0.076	+0.076

Table 4: Winner by forward target. Volatility wins the magnitude target; shape metrics win every shape-sensitive target.

Forward target	vol rank corr (range)	shape beats vol
max drawdown (magnitude)	+0.36 to +0.62	none, any bed
−skew (shape)	−0.21 to −0.07	most, every bed
downside fraction (shape)	−0.25 to −0.05	most, every bed
tail-exceedance freq (shape)	−0.52 to −0.10	most, every bed

is high when upside dominates downside, so its zero increment beyond volatility reflects a real null, not a wrong sign.

5 WHY VOLATILITY WINS, AND CAN ANYTHING BEAT IT?

5.1 Persistence

Volatility wins because it is persistent, the founding fact of ARCH/GARCH [7, 12]. Trailing volatility forecasts forward volatility (rank correlation 0.42–0.84), which in turn drives forward drawdown. Table 5 decomposes this chain by bed. Both links exceed the direct trailing-vol→drawdown rank correlation they compose. We resist calling forward volatility a *sufficient statistic*. A naive same-window mediation suggests it, but that test is

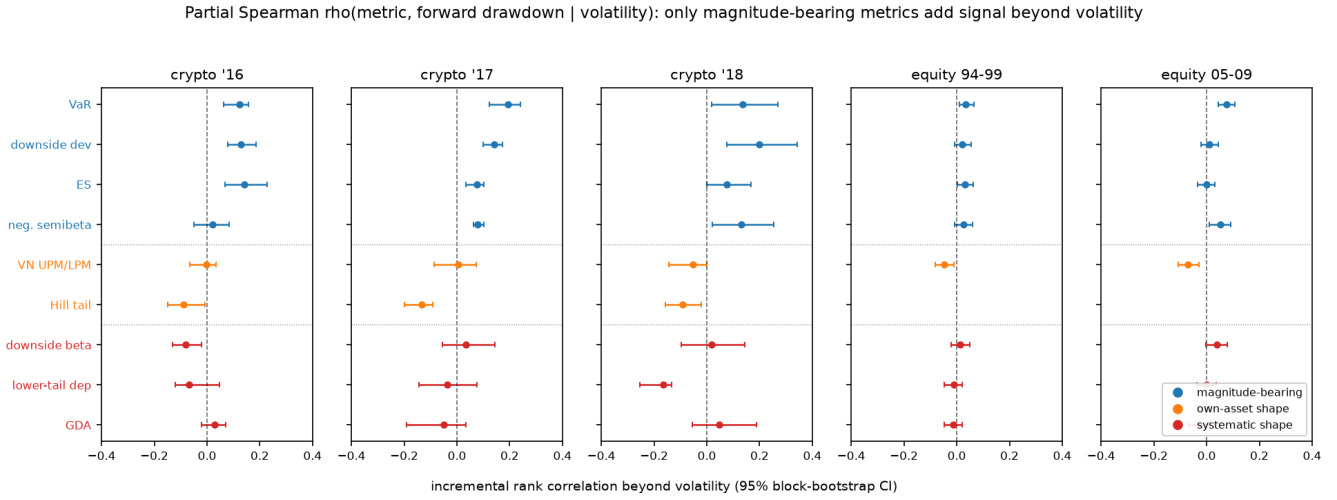


Figure 1: Incremental signal beyond volatility, by bed. Partial Spearman $\rho(\text{metric, drawdown} \mid \text{volatility})$, grouped by whether a metric embeds downside magnitude. Only the magnitude-bearing metrics (blue: VaR, downside deviation, ES, and negative semibeta, whose down-down semicovariance construction carries downside magnitude) sit reliably right of zero. The magnitude-free shape metrics add ≈ 0 or subtract, both the own-asset ones (orange: the Viole–Nawrocki (VN) UPM/LPM ratio and the Hill tail exponent) and the systematic co-movement ones (red: downside beta, lower-tail dependence, and the disappointment-aversion (GDA) proxy). Magnitude beats shape on both sides of the own-asset/systematic divide. Whiskers are 95% block-bootstrap CIs; dotted lines separate the three groups.

Table 5: Persistence chain (mean per-date Spearman; dd = drawdown; all CIs exclude 0).

Bed	trail→fwd vol	fwd vol→dd	(trail→dd)
crypto-16	0.68	0.77	0.62
crypto-17	0.42	0.62	0.36
crypto-18	0.42	0.53	0.38
equity 94	0.84	0.71	0.61
equity 05	0.80	0.70	0.56

contaminated (forward volatility and drawdown share a window). Measured on a disjoint, predictive window (forward volatility over $[t, t+30]$ and drawdown over $[t+30, t+120]$, so mediator and outcome share no days), trailing volatility retains a partial correlation of 0.32–0.34 on equity (Figure 2). Because that disjoint mediator is a noisy 30-day proxy, we read this as consistent with incomplete mediation rather than proof, so persistence is a leading but incomplete explanation, not a mechanism that reduces drawdown to one number.

Horizon dynamics. Persistence and forecastability rise and fall together. As we sweep the forecast horizon from 14 to 90 days, volatility persistence and the drawdown rank correlation move in step (Figure 3). On equity both rise (persistence $0.66 \rightarrow 0.84$, rank correlation $0.47 \rightarrow 0.61$) at a roughly constant gap, while on crypto both are flat. Wherever persistence is higher, so is the rank correlation, consistent with persistence driving most of the forecastable variation and the constant gap being the part it does not explain.

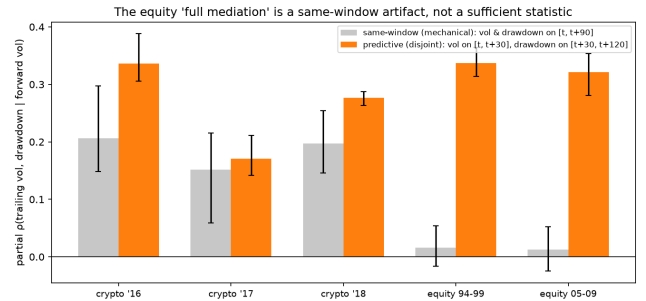


Figure 2: Mediation: partial $\rho(\text{trailing vol, drawdown} \mid \text{forward vol})$. The same-window (mechanical) design gives ≈ 0 on equity, falsely suggesting full mediation. The predictive (disjoint-window) design gives 0.32–0.34, so forward volatility does not fully screen off trailing volatility.

5.2 Does the edge pay?

A rank correlation is not a profit-and-loss statement. The magnitude-not-shape split of §4, volatility plus a single tail-magnitude term, implies one natural composite, the equal-weight sum $z(\text{vol}) + z(\text{VaR})$ with no fitting, where $z(\cdot)$ is the cross-sectional z-score (each metric standardized across the assets on a given date). VaR was fixed as the composite’s second term before the locked panel was opened. It was the only metric whose increment survived correction on all five exploratory beds, and the weights are untuned, so the locked panel is fully out-of-sample. There it shows no significant out-of-sample gain over volatility (Newey–West $t = 0.67$, lag 3), beating it only on fat-tailed

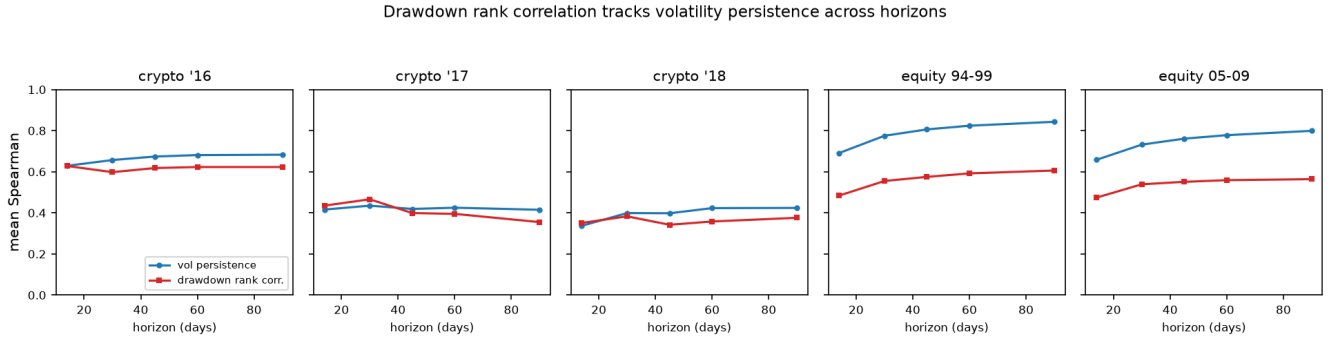


Figure 3: Volatility persistence (blue) and drawdown-forecast rank correlation (red) versus forecast horizon (14–90 days), by bed. The two curves rise and fall together at a roughly constant gap, so where volatility is more persistent it also forecasts drawdown better.

Table 6: Composite $z(\text{vol}) + z(\text{VaR})$ vs. volatility, per-date Spearman and difference (95% CI). The difference is significantly positive only on crypto-bull and significantly negative, though trivially small, on equity-94; n.s. = not significant (the difference’s 95% CI includes 0). The locked panel trains on 2017 and is scored once on the 2018 crash.

Test bed	vol ρ	comp ρ	difference (95% CI)
crypto-16	0.62	0.62	−0.00 (n.s.)
crypto-17	0.36	0.40	+0.05 [0.02, 0.07]
crypto-18	0.38	0.39	+0.01 (n.s.)
equity 94	0.61	0.60	−0.008 [−0.011, −0.004]
equity 05	0.56	0.56	−0.00 (n.s.)
Locked panel: 0.376 → 0.388, Newey–West $t = 0.67$ (n.s.).			

crypto-bull (Table 6). That crypto-bull edge is +0.05, below the bed’s 80%-power MDE of 0.09, so like VaR’s crypto claims it is suggestive rather than powered. A cost-aware tercile backtest agrees (Figure 4). The high-risk tercile realizes 10–30 percentage points more forward drawdown than the low-risk tercile on every bed, net of a 20-basis-point (bps) cost charged on portfolio turnover at each rebalance (terciles, rebalanced per formation date), and the composite adds nothing net of turnover except, again, on crypto-bull. Volatility is the workhorse. Magnitude adds a little, and only where tails are fat.

5.3 Can a machine find more?

If the residual is this small, a tireless automated search should fail out-of-sample and merely overfit validation. We test this on a strict train/validation/locked-test split, the locked test scored once after the metric is frozen. On the original locked panel, an LLM agent rewriting a scoring function and a 20,000-configuration sweep both inflate the best *validation* score with search intensity while the *locked-test* score plateaus at the volatility baseline (Table 7, Figure 5). Of the swept configurations that beat volatility on validation, only 43% also beat it on the locked test, despite a 0.94 validation–test rank correlation.

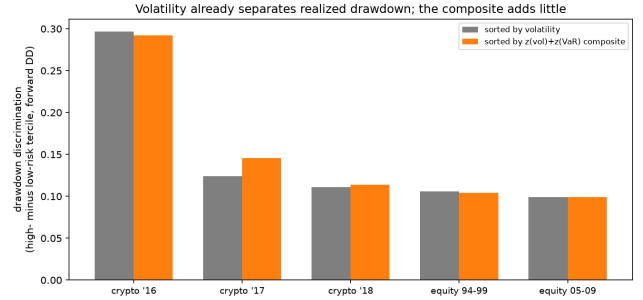


Figure 4: Realized drawdown discrimination (high minus low-risk tercile) by bed, sorting on volatility vs. the composite. Volatility already separates drawdown strongly; the composite is indistinguishable except marginally on crypto-bull.

The discovery loop. The agent is Claude Opus 4.8. It is given a frozen scorer it may not edit, a train/validation split, and a single file defining `build_scores(panel)`. Each iteration it reads the log of past attempts and the current metric, rewrites the metric with one new idea (any Python over the per-row features volatility, ES, VaR, LPM₂, downside deviation, trailing drawdown, skew, kurtosis, and market cap, fitting any parameters on the train split only), scores it on validation, and keeps the edit if validation improved, otherwise reverts. It never sees the locked test, which a separate frozen script scores exactly once after the metric is finalized. The run we report ran 51 iterations (14 kept, 37 discarded), lifting validation from 0.31 to 0.34. The 20,000-configuration sweep enumerates the same feature-combination space (sums, ratios, products, gates, interactions) exhaustively, and the general nonlinear generator samples 1,500 such candidates per run. All three optimize the same validation score against the same locked test.

This is not a single-panel artifact. We re-ran a general nonlinear candidate generator (1,500 candidates per run) across a grid of holdout panels $\times$ 3 search seeds in two families: *out-of-time*, a fixed-length test window slid through the timeline with validation always chronologically earlier; and *random*, order-ignoring

Table 7: Best-of- N discovery on the original locked panel. Validation inflates with search intensity; the locked test does not (volatility baseline 0.376). Validation and test are different splits, so compare trends, not levels.

Search intensity N	Best validation	Its locked test
1	0.27	0.31
100	0.35	0.37
20,000	0.356	0.373

repartitions of the non-train dates. The validation-selected metric’s locked-test edge over volatility (test gap) is small but positive, averaging +0.020 out-of-time (clustered $t = 4.3$ across the six panels) and +0.035 under random repartitions ($t = 5.3$), while the validation–test rank correlation stays 0.94–0.99 throughout. Two readings follow. First, the out-of-time edge is tiny and consistent with search rediscovering the same tail-magnitude increment of \$4, not new signal. Second, the random-split edge is the larger of the two, a signature of temporal leakage, because order-ignoring repartitions interleave validation and test dates and let volatility’s persistence leak across the split, manufacturing a phantom edge. Only chronological holdouts are honest in persistent panels. Either way the edge is economically negligible, and the non-LLM sweep reproduces it, so the danger is the cheap search, not the language model.

The statistics are classical data snooping [4, 28]. What is new is that an autonomous agent removes the last brake on trial count, namely analyst effort, so a locked, survivorship-free holdout scored exactly once becomes mandatory, not merely advisable.

5.4 What is volatility for?

The clearest positive use is forecasting outright blow-ups. On the locked crypto test, volatility predicts which assets crater (forward return $\leq -80\%$) with ROC-AUC 0.68 (0.95 for the stricter $\leq -90\%$ cutoff) and a 2.5 $\times$ top-decile lift, while a gradient-boosting model over all ten metrics, chosen on validation (AUC 0.73), overfits and loses on the locked test (AUC 0.59). The blow-up base rate is only $\approx 1.2\%$, so these AUCs are noisy and we read them as ordinal. Again the size, not the shape, of recent moves carries the signal. This is a ranking result, not a deployment recommendation (cost, liquidity, and reflexivity frictions are unmodeled).

6 HOW MUCH DO THE DISCIPLINES MATTER?

The three fair-field disciplines change what one would report, not the verdict. We ask which shape metric one would have credited without each.

Without survivorship-free data. We rebuild the crypto universe survivors-only, keeping at each formation date only coins that neither delisted nor ended below 5% of their all-time peak, about a third of the point-in-time top 100. On the 2018 crash bed this erases shape’s penalty and inflates the magnitude edge (Table 8). Negative semibeta, a co-crash shape metric, flips from significantly worse than volatility to a tie, downside beta closes its gap, and VaR’s partial rises. This happens precisely in the crash regime, where

Table 8: Survivorship ablation on the crypto-2018 crash bed, vs. volatility (block-bootstrap 95% CIs where estimated). Dropping the coins that died turns a co-crash shape metric’s loss into a tie and inflates the magnitude edge, yet no shape metric beats volatility either way.

crypto-2018	full universe	survivors-only
neg. semibeta (h2h)	−0.067 [−0.131, −0.003]	+0.055 [−0.082, 0.178]
downside beta (h2h)	−0.23	−0.01
VaR (partial ρ)	+0.135	+0.205

hiding the coins that died most reshapes the cross-section. It does not manufacture a winner. No shape metric beats volatility even survivors-only.

Without multiple-testing correction. Across the ~ 45 head-to-head cells the only metrics nominally beating volatility at $p < 0.05$ are downside deviation and VaR, both magnitude measures, and only on crypto-bull. No shape metric is even nominally positive, so Benjamini–Hochberg is not what holds shape back.

Without a locked holdout. The discovery experiment (§5, Table 7 and Figure 5) makes this concrete. A validation-selected metric’s inflated validation score reads as a win, while its once-scored locked-test score sits at the volatility baseline. Even stripped of every guardrail, no shape metric beats volatility.

7 DISCUSSION AND CONCLUSION

The pieces form one coherent picture. Volatility sets a high bar because it is persistent (§5). The only thing left to add is the magnitude of tail losses, and only a little, mostly in fat-tailed markets (§4). So no standalone metric beats volatility (§3), a pre-registered composite yields no significant out-of-sample gain, automated search cannot find one, and the economics track the statistics (§5). The recurring failure of magnitude-free asymmetry and shape metrics (including the partial-moment ratio [25]) is not an implementation accident. Once symmetric volatility is accounted for, asymmetric structure adds incremental cross-sectional information for forecasting drawdown only when it also carries downside magnitude, as in negative semibeta; pure shape adds nothing. The fair-field disciplines change what one would report, not the verdict (§6). Survivorship bias alone flips a co-crash shape metric from significantly worse than volatility to a tie and inflates the apparent magnitude edge, yet the verdict that no shape metric beats volatility is robust to dropping any single discipline. The disciplines guard the honesty of the numbers, not the direction of the conclusion.

We are deliberately careful about the strength of the persistence story. An earlier version of this study claimed forward volatility was a sufficient statistic, found that claim was a same-window measurement artifact, and corrected it in this version, just as an earlier “survivorship masks downside value” thesis was dropped when a hardened test refuted it. The paper reports only claims that survived stress-testing.

The practical takeaway is plain but firm. For cross-sectional drawdown forecasting, use volatility (optionally with a small tail-magnitude term in fat-tailed markets) and treat any downside metric “discovered” to beat it, by hand or by machine, as unproven

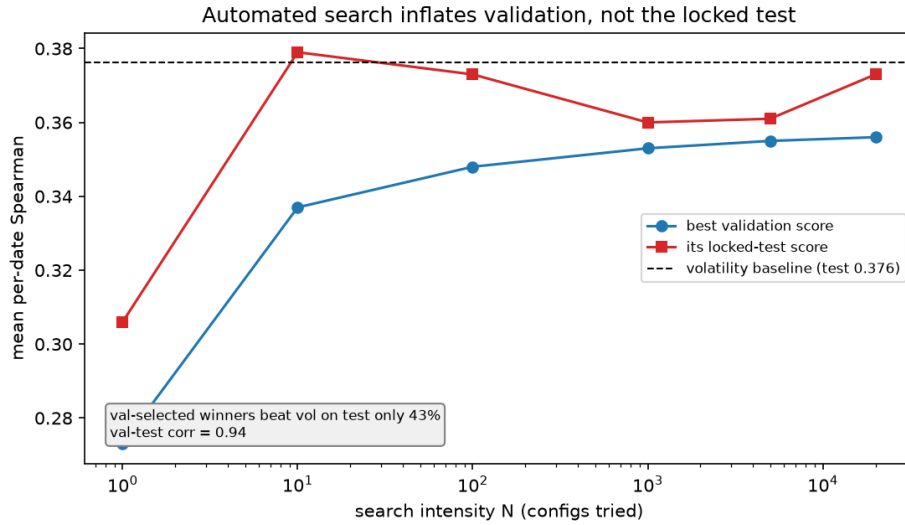


Figure 5: Automated discovery does not beat volatility out of sample by more than a sliver. As search intensity grows, the best validation score climbs while its locked-test score stays flat at the volatility baseline. On this (2017 → 2018-crash) holdout, only 43% of the configurations that beat volatility on validation also beat it on the locked test. Across 12 panels × 3 seeds the locked-test edge is statistically positive but economically negligible (+0.02 out-of-time), and larger under order-ignoring random splits, the signature of temporal leakage.

until it clears a locked, survivorship-free holdout. On a level field, volatility is not glamorous, but it is very hard to beat.

Limitations. The performance measure is rank correlation, not a full tradable P&L (\$5 is a first economic check, not a portfolio study); the crypto beds have few formation dates (7–9), so their significance is suggestive and the null “shape adds nothing” leans on the well-powered equity beds; the trailing window is fixed at 180 days and the horizon sweep varies only the forward side, so sensitivity to the trailing length is unexamined; and the repurposed pricing metrics are daily adaptations, so results speak to drawdown forecasting, not the originals’ pricing claims.

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